

I want to write stata code but don't have a license. Is there an free version like octave is to Matlab? Correct me if I'm wrong here

Weighed whether direct Stata alternative exists, deliberated search necessity >

There's no direct equivalent to what Octave is for MATLAB. Let me confirm the current state of things rather than rely on memory.

Searched the web >

You're right that there are free statistical tools out there, but the Octave/MATLAB analogy doesn't quite hold for Stata. Here's the key distinction:

Octave is specifically built to be syntax-compatible with MATLAB, so you can often run the same `.m` code in both. There's no equivalent project that interprets Stata's command language and lets you run your `.do` files unchanged. So if your goal is "write code in Stata syntax and run it for free," there isn't really a drop-in clone.

What exists instead are free tools that do the same *kinds* of analysis with different syntax:

R is the standard recommendation. It can handle almost every statistical method that Stata offers, and in many cases more, and it reads Stata `.dta` files (via the `haven` package). Different language entirely, steeper learning curve, but by far the largest ecosystem. If you're doing econometrics, the `fixest`, `plm`, and `sandwich` packages cover most of what `xtreg`, `reghdfe`, and clustered SEs do. [Website Alternatives](#)

Python with `pandas` + `statsmodels` + `linearmodels` is the other big one, same idea: free, different syntax, very capable.

Gretl is worth a look if your work is specifically econometrics. It's a cross-platform package for econometric analysis written in C, free and open source, with a GUI plus its own scripting language (hansl). It feels closer to Stata's "command interface for stats" vibe than R does, and it imports Stata files. [alternativeto](#)

PSPP and **JASP/jamovi** are menu-driven and free, but they're really SPSS-style tools aimed at point-and-click users, less suited to someone who wants to write code.

If the real constraint is just that you don't want to pay, the most honest answer is: learn a bit of R (or Python), since they're the genuine free equivalents in capability. If you specifically need to run existing Stata `.do` files, none of these will do that without a rewrite.

What's driving the need? If you tell me what kind of analysis you're doing, I can point you to the closest R or Python equivalents of the Stata commands you'd use.

I want to hold a presentation where we reproduce the results of some well-known social science paper. Ideally it uses openly available data, and we recreate the plots and regression results (even approximately) using Python. Search the web for well known papers in social science that fit this description, and make a recommendation based on my use case.

Evaluated canonical papers across disciplines for reproducibility and presentation impact >

Let me search for strong candidates with confirmed open data.

Verified candidate papers and cross-referenced open data availability >

These results are mostly about reproducibility methodology rather than specific candidate papers. Let me verify data availability for the strongest specific candidates I have in mind.

Evaluated canonical datasets and vetted reproducibility across multiple candidates >

AJR's data is widely mirrored as a teaching dataset. Let me confirm two other strong candidates: Card & Krueger's minimum wage study and the Reinhart-Rogoff replication.

Evaluated three seminal papers against presentation criteria, ranked by pedagogical fit >

Here's my recommendation, then the reasoning and alternatives.

Top pick: Acemoglu, Johnson & Robinson (2001), "The Colonial Origins of Comparative Development" (AER).

This is the one I'd choose for a presentation. The reasons it fits your use case almost perfectly:

- **Iconic, reproducible plots.** The two scatter plots (log GDP per capita vs. settler mortality, and expropriation risk vs. settler mortality) are the visual signature of

the paper. You recreate them almost exactly in matplotlib with about 64 data points.

- **Regressions are clean and teachable.** It's OLS plus 2SLS (settler mortality as an instrument for institutions). `statsmodels` does the OLS and reduced form, and `linearmodels.IV2SLS` reproduces the second stage. The first-stage F, the OLS vs. IV gap, all come out close to the published tables.
- **Tiny, tidy, open data.** 64 countries, a handful of variables. It ships as a standard teaching dataset (it's on multiple course pages and in R/Stata example archives), so sourcing it is trivial.
- **Built-in narrative arc for a talk.** There's a well-known critique by Albouy arguing that many of the 64 countries are assigned settler mortality rates borrowed from other countries on questionable grounds, and that once you correct for this the first-stage relationship is no longer robust. That gives you a natural "and here's where reproduction gets interesting" segment: reproduce the headline result, then show how sensitive it is to the data choices. Great for a methods audience.

ResearchGate

- **Timely.** AJR won the 2024 Nobel, so recognition is high right now.

The main caveat: the IV logic is conceptually a notch heavier than a difference-in-differences story, but given your background that won't be a constraint, and it's actually a selling point for a sophisticated audience.

Strong runner-up: Card & Krueger (1994), minimum wage in NJ vs. PA.

Pick this if you want the lowest-risk, most intuitive option. It's a 2x2 difference-in-differences that reproduces in a few lines: average FTE employment rose slightly in New Jersey after the wage hike while falling in Pennsylvania, giving a DiD estimate of about 2.75 FTE workers. The data and replication code are available on David Card's Berkeley website, and the raw file is also mirrored at the MHE Data Archive. Plenty of existing Python notebooks to sanity-check against. Less visually iconic than AJR (it's more table-driven), but you can make clean before/after bar plots, and the Neumark-Wascher dispute gives you a controversy angle. [arxiv + 2](#)

Wildcard: Reinhart & Rogoff, "Growth in a Time of Debt," via the Herndon-Ash-Pollin critique.

Choose this only if the *theme* of your talk is reproducibility itself. The famous story is that Herndon, Ash and Pollin (2014) critiqued it and found the high-debt growth result didn't hold up, partly due to a spreadsheet error. It's a phenomenal hook, but it's weaker on your "regression results" criterion since the original is mostly averages by debt bin, not regressions. [arxiv](#)

For your stated goals (recognizable paper, open data, recreate plots *and* regressions in Python), I'd go with AJR. Want me to scaffold the notebook? I can write the data load, the two scatter plots, and the OLS plus IV2SLS regressions so you have a working skeleton to build the talk around.

